

Hoai-Chau Tran

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC)

Illinois, USA

Ph.D. Student (Expected day of graduation: 5/2030)

8/2025-present

- **Research interest:** Optimization for efficient 3D Transformers and multimodal models including (VLM, VLA), kernel development for efficient attention mechanisms, quantization and compression techniques, long-range sequence learning, and embodied AI systems.
- **GPA:** 4.0/4.0.

University of Science, VNU-HCM

Ho Chi Minh City, Vietnam

Bachelor of Science, Advanced Program in Computer Science

2018-2022

- **GPA:** 8.9/10.0.

SELECTED PUBLICATIONS

List of publications on [Google Scholar](#).

1. Tuan Anh Tran, Duy Minh Ho Nguyen, **Hoai-Chau Tran**, Michael Barz, Khoa D Doan, Roger Wattenhofer, Vien Anh Ngo, Mathias Niepert, Daniel Sonntag, Paul Swoboda. **How Many Tokens Do 3D Point Cloud Transformer Architectures Really Need?**. *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*..
2. **Hoai-Chau Tran**, Duy M. H. Nguyen, Duy M. Nguyen, Trung-Tin Nguyen, Ngan Le, Pengtao Xie, Daniel Sonntag, James Y. Zou, Binh T. Nguyen, Mathias Niepert. **Accelerating Transformers with Spectrum Preserving Token Merging**. *Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*..
3. **Hoai-Chau Tran**, Anh-Duy Tran, Kim-Hung Le. **DetectVul: A statement-level code vulnerability detection for Python**. *Future Generation Computer Systems (FGCS)*. 163 (2025) p. 107504. 2025.
4. **Hoai-Chau Tran**, Chi H Nguyen, Duy Minh Ho Nguyen, Mathias Niepert, Fan Lai, Khoa D Doan **Spars-eSAM: Structured Sparsification of Activations in Segment Anything Models**. *In Submission*.

RESEARCH EXPERIENCE AND PROJECTS

SparseSAM

Illinois, USA

Research Assistant

8/2025-present

- Designed SparseSAM, a training-free framework that accelerates SAM encoders by jointly sparsifying attention and MLP layers while maintaining token identity.
- Introduced Stripe-Sort Attention and Residual-Consistency MLP to eliminate dynamic mask overhead and route only key tokens through MLP layers.
- Achieved up to 2.05× throughput speedup and 2.8× memory reduction on A100/RTX 3090 GPUs across HQ-44K and ImageNet benchmarks with minimal accuracy loss.
- **Advisors:** Prof. Khoa D. Doan, Prof. Fan Lai

MyGPT2: Custom CUDA Implementation for LLM Inference

Illinois, USA

Independent Research

9/2025-11/2025

- Implemented GPT-2 inference entirely in *custom CUDA kernels* (attention, matrix multiplication, *quantization*) from scratch without relying on high-level frameworks like PyTorch or cuBLAS.
- Designed and optimized *FlashAttention kernels* using advanced techniques including shared-memory tiling, Split-K decomposition, local attention windows, and customized Tensor Core instructions, achieving 75% throughput and memory-efficiency compared to the flash-attn package.
- Integrated *Activation-Aware Quantization (AWQ)* with calibration, per-channel scaling, and inference-time dequantization to substantially reduce memory bandwidth usage while preserving model accuracy.

VinUni-Illinois Smart Health Center, VinUniversity

Ha Noi, Vietnam

Research Assistant

8/2024-7/2025

- **Description:** Research on *3D Transformer architectures* (PTv3, Sonata, etc.) and development of *Git-Merge3D*, a token compressing algorithm for *accelerating 3D point cloud transformers*. The method estimates voxel token importance from spatial structures and selectively merges low-importance tokens, enabling up to 90% token reduction while preserving accuracy. Achieved substantial efficiency gains by reducing PTv3 computation from 107.5 GFLOPs to 19.9 GFLOPs and memory footprint from 10.12 GB to 1.6 GB, with negligible mIoU degradation (e.g., 77.6 → 77.4).

- **Description:** Collect data and developed **DetectVul**—first statement-level Python vulnerability detection model without graph extraction; achieved +25.45% F1 improvement over GCN and +18.05% over GAT across 211K statements. Released full pipeline, datasets, and model artifacts for reproducibility.
- **Collaborators:** Anh-Duy Tran, Kim Hung Le
- **Advisor:** Prof. Kim Hung Le

INDUSTRY EXPERIENCE

- Developed and maintained REST APIs for personal finance management, including budgets, expenses, and transaction histories.
- Created tools and scripts to assist other teams in managing development databases and benchmarking services’s performance and resilience.
- Integrated SQLite for local data storage to enable faster transaction history query performance on mobile devices.

- Developed a quantization-aware distillation pipeline to compress vision models to 4-bit precision, reducing model size to 5 MB for deployment on edge NPUs while preserving accuracy.
- Performed model surgery and graph optimization to ensure compatibility with the NPU compiler and successfully deploy models on NPU hardware.

SKILLS

1. **Programming languages:** Python, C/C++, CUDA, Java, SQL.
2. **Technologies:** PyTorch, CUDA, Numpy, HuggingFace, Git, Scikit-learn, AWS, Docker, Linux, L^AT_EX.
3. **Languages:** English (*fluent*), Vietnamese (*native*).

RELEVANT COURSES

- **CS598 - System For Gen AI** by Prof. Fan Lai (UIUC - Spring 2026)
- **CS445 - Computational Photography** by Prof. Yuxiong Wang (UIUC - Spring 2026)
- **CS446 - Machine Learning** by Prof. Liangyan Gui (UIUC - Fall 2025)
- **CS483 - Applied Parallel Computing** by Prof. Volodymyr Kindratenko (UIUC - Fall 2025)
- **Deep Learning Specialization** by Prof. Andrew Ng (Coursera)
- **Linear Algebra** by Prof. Nguyen Huu Anh (University of Science - Summer 2019)
- **Probability and Statistics** * by Prof. Luu Quoc Toan (University of Science - Fall 2020)
- **Calculus I,II,III** by Prof. Le Van Luyen (University of Science - Fall 2019, Spring 2020, Fall 2021)

PROFESSIONAL SERVICES

1. **Reviewer:** NeurIPS (2025, 2026), ICLR (2025, 2026), ICML (2026)
2. **Volunteer:** NeurIPS (2025)
3. **Workshop committee:** Resource-Adaptive Foundation Model Inference (AdaptFM, ICML 2026)

REFERENCES

1. **Prof. Fan Lai**, Assistant Professor at the Siebel School of Computing and Data Science - University of Illinois at Urbana-Champaign (UIUC), USA fanlai@illinois.edu
2. **Prof. Khoa D Doan**, Associate Director at the VinUni-Illinois Smart Health Center. Assistant Prof. at College of Engineering & Computer Science, VinUniversity, Vietnam khoa.dd@vinuni.edu.vn
3. **Babak Rasolzadeh**, Sr. ML Manager, MIND team, Apple Inc, USA. b_rasolzadeh@apple.com
4. **Hamid Nejatollahi**, Research Scientist, MIND team, Apple Inc, USA. h_nejatollahi@apple.com

*Obtained A+ for excellent performance.